· CMR TECHNICAL CAMPUS

· UGC AUTONOMOUS

· Accredited by NBA & NAAC with 'A' Grade

· Approved by AICTE, New Delhi and JNTU Hyderabad

· Department of Computer Science and Engineering

TITLE

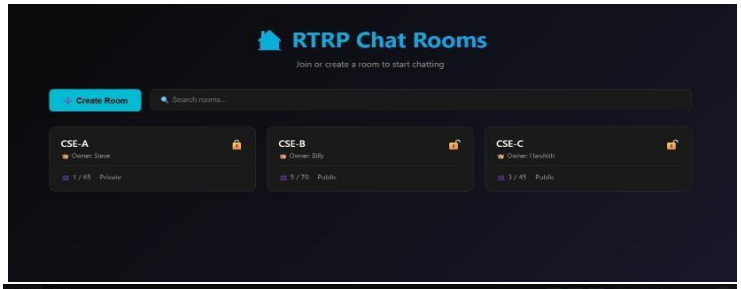
ABSTRACT

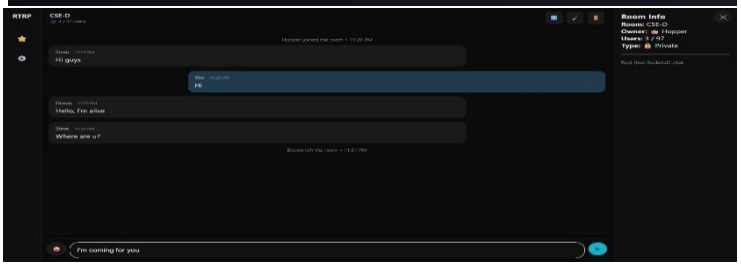
Open Chat is a real-time, room-based chat application built with Node.js, Express, and Socket.IO, enabling seamless multi-user communication through persistent WebSocket connections. Users can create and join public or private (password-protected) chat rooms, with features like typing indicators, live user counts, and instant join/leave notifications. The application follows an event-driven architecture with server-side room management, input validation, and controlled user limits to ensure stability and basic security. Designed with a responsive dark-themed UI using glassmorphism effects, it delivers a smooth experience across desktop and mobile devices without requiring any database or persistent storage. Deployed on Render via GitHub integration, this project demonstrates practical proficiency in real-time web systems, responsive design, and cloud deployment workflows.

DESCRIPTION

Open Chat is a real-time chat application I built using Node.js, Express, and Socket.IO, where users can create and jump into public or private password protected rooms instantly — no page reloads, no database, just seamless live communication. It comes packed with features like typing indicators, live user counts, instant join/leave notifications, and automatic room cleanup when everyone leaves.

The interface has a clean dark theme with glassmorphism effects that looks great on any device, from mobile to desktop. I deployed it on Render through GitHub, and the whole project gave me solid hands-on experience with WebSockets, event-driven architecture, responsive design, and cloud deployment.





EXISTING SOLUTIONS

1. **Slack**
2. **Discord**
3. **Microsoft Teams**
4. **Rocket.Chat**
5. **Mattermost**

CONCLUSION

Open Chat turned out to be a really rewarding project that brought together everything from real-time WebSocket communication to responsive UI design and cloud deployment — all under one roof. Building it with Node.js, Express, and Socket.IO showed just how powerful and approachable real-time web development can be, even without a database. The app handles room creation, private access, live messaging, and automatic cleanup in a clean and efficient way that genuinely works well in practice. There's still a lot of exciting room to grow — things like message history, user authentication, and direct messaging are natural next steps that could take this project even further.

Batch-15:

K. VISHNU TEJA 247R1A0595
P. HARSHITH 247R1A05B0
P. GNYANESHWAR 247R1A05B2

FACULTY SUPERVISOR
Dr V Naresh Kumar
Associative Professor.